

Wireless Access Electromagnetics

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Assignment 04

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Problem 1:

Design of a Rectangular Patch Antenna at 5.8 GHz

We are asked to design a rectangular patch antenna with a resonance frequency of 5.8 GHz, with an antenna input impedance constraint of $Z_{in}(f_0) \leq 120$ ohms. I used the provided grad script to find the fraction W/λ_0 using Matlab, and consequently calculated the width and length of the patch. The solution is given below:

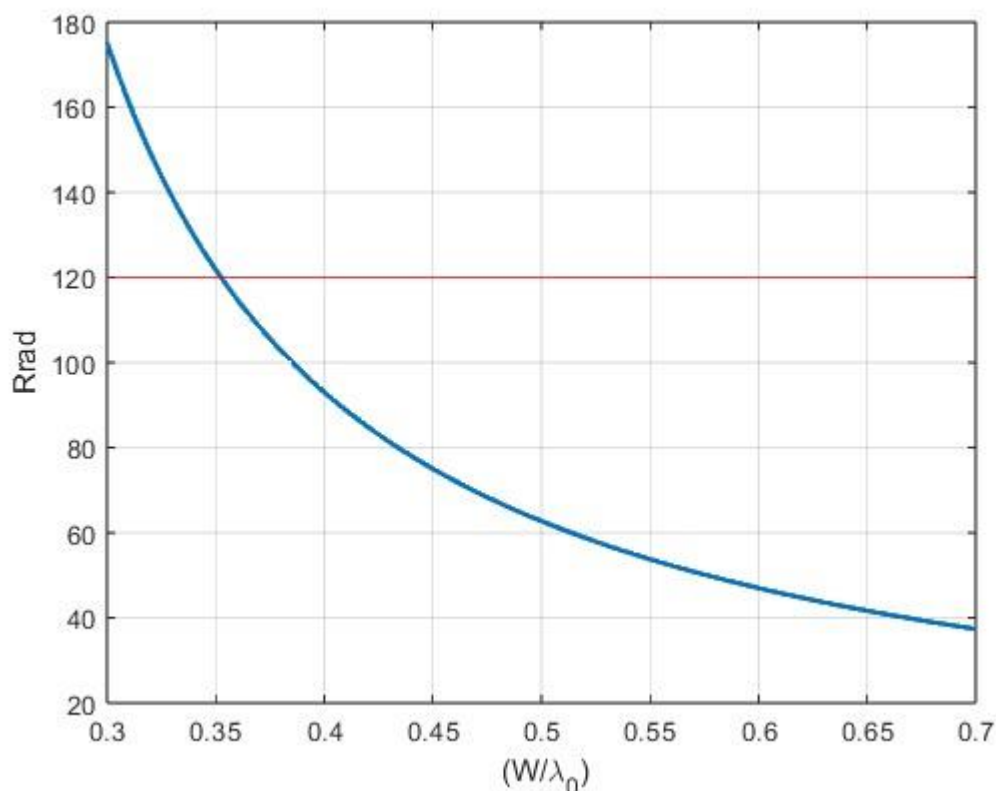


Figure 1

Ex 1,

Given:

$$\epsilon_r = 4.45$$

$$h = 1.55 \text{ mm}$$

$$f_0 = 5.8 \text{ GHz}$$

Initially, we find the ratio $l/\lambda_0 = \frac{1}{2} \times \frac{1}{\sqrt{\epsilon_r}} = \frac{1}{2\sqrt{4.45}} \approx 0.237$

$$\lambda_0 = \frac{c}{f_0} = \frac{3 \times 10^8}{5.8 \times 10^9} \approx 5.172 \text{ cm}$$

To find W/λ_0 , we use the Grad patch polyapp that is provided and iterate through the range for W and l . I found $\frac{W}{\lambda_0}$ to be $\approx 0.353 \Rightarrow W = 0.353 \times 5.172 = 1.825 \text{ cm}$

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{W} \right)^{-1/2} \approx 3.94$$

$$l_{\text{eff}} = \frac{l_g}{2}, \quad l_g = \frac{\lambda_0}{\sqrt{\epsilon_{\text{eff}}}} = \frac{5.172}{\sqrt{3.94}} \approx 2.6 \text{ cm} \Rightarrow l_{\text{eff}} = 1.3 \text{ cm}$$

$$\Delta L = 0.412 \cdot h \cdot \frac{(\epsilon_{\text{eff}} + 0.3) \left(\frac{W}{h} + 0.264 \right)}{(\epsilon_{\text{eff}} - 0.258) \left(\frac{W}{h} + 0.8 \right)} \approx 0.07 \text{ cm}$$

$$\text{Now, for } l = l_{\text{eff}} - 2\Delta L = 1.3 - 2(0.07) = 1.162 \text{ cm}$$

Design summary: $\frac{W}{\lambda_0} = 0.353$

$$\frac{l}{\lambda_0} = \frac{1.162}{5.172} = 0.225 \Rightarrow R_{\text{in}} = 119.6 \leq 120 \Omega$$

Using the provided Grad patch script, I found W/λ_0 to be 0.353, and consequently, $W = 18.25 \text{ mm}$ and $L = 11.62 \text{ mm}$.

Problem 2

Bandwidth analysis of Antenna design of problem 1

Part 1:

Ex 2, Part 1,

$$Z_{\infty} = \frac{120 \Omega}{\sqrt{\epsilon_{eff}} \left[\frac{W}{h} + 1.393 + 0.667 \ln \left(\frac{W}{h} + 1.44 \right) \right]} \quad \text{for } \frac{W}{h} \geq 1$$

$$\Rightarrow \frac{120 \Omega}{\sqrt{3.94} [11.77 + 1.393 + 0.667 \ln(11.77 + 1.44)]}$$

$$Z_{\infty} \approx 12.76 \Omega$$

$$G_{edge} = \frac{1}{(2 \times R_{in})} = \frac{1}{2 \times 119.6} \approx 0.00425$$

$$R_{edge} = 1/G_{edge} = 237.4 \Omega$$

$$\Gamma_B = \frac{1/G_{edge} - Z_{\infty}}{1/G_{edge} + Z_{\infty}} = \frac{237.35 - 12.76}{237.35 + 12.76} \approx 0.898$$

$$\Gamma_A^+ = \Gamma_B^- \exp(-2j k(f) L_{eff}) \quad ; \quad k(f) = \frac{2\pi}{\lambda} = \frac{2\pi}{\frac{c}{f\sqrt{\epsilon_r}}}$$

$$\Rightarrow 0.89 \exp \left(-2j \frac{2\pi}{\frac{1}{\sqrt{\epsilon_r}}} \times \frac{10}{2\sqrt{\epsilon_r}} \right) \quad L_{eff} = 10/2\sqrt{\epsilon_r}$$

$$\Rightarrow \Gamma_A^+ = 0.89 e^{-2j\pi f/f_0} \quad ; \quad f/f_0 [0.95 f_0, 1.05 f_0]$$

$$Z_{in}(f) = Z_{\infty} \frac{1 + \Gamma_A^+(f)}{1 - \Gamma_A^+(f)} \parallel \frac{1}{G_{edge}}$$

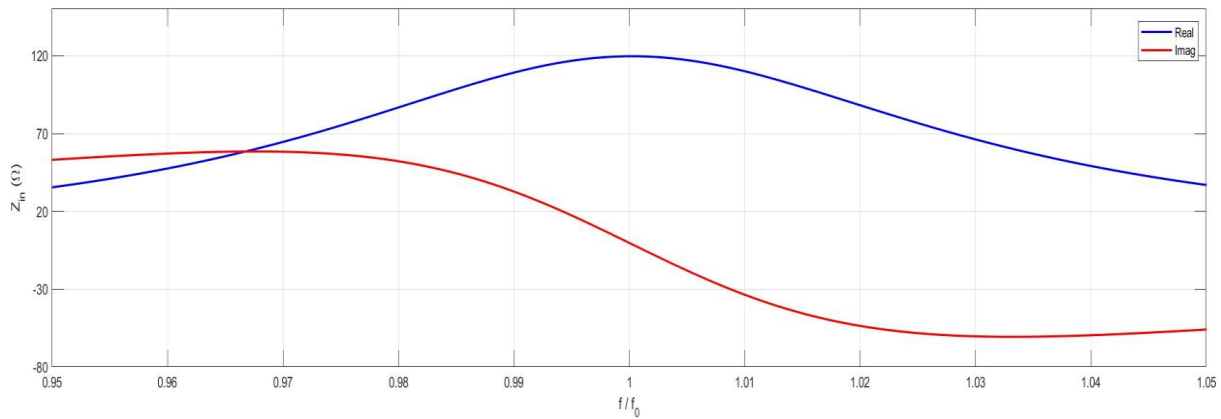


Figure 2: Real and Imaginary Parts of Z_{in}

Part 2:

Part 2.

$$\Gamma_{in}(f) = \frac{Z_{in}(f) - Z_0}{Z_{in}(f) + Z_0}$$

$$= \left| \frac{Z_{in}(f) - 120}{Z_{in}(f) + 120} \right|$$

At 5.8 GHz (design frequency), $\Gamma_{in}(f) \approx 0$.

$$|\Gamma_{in}| \leq \frac{VSWR - 1}{VSWR + 1} = \frac{1}{3} \Rightarrow \text{Return loss} \approx \boxed{-9.54 \text{ dB}}$$

The absolute bandwidth was found to be 145 MHz.

$$\text{Relative bandwidth} = \frac{145}{5800} \approx \boxed{2.5\%}$$

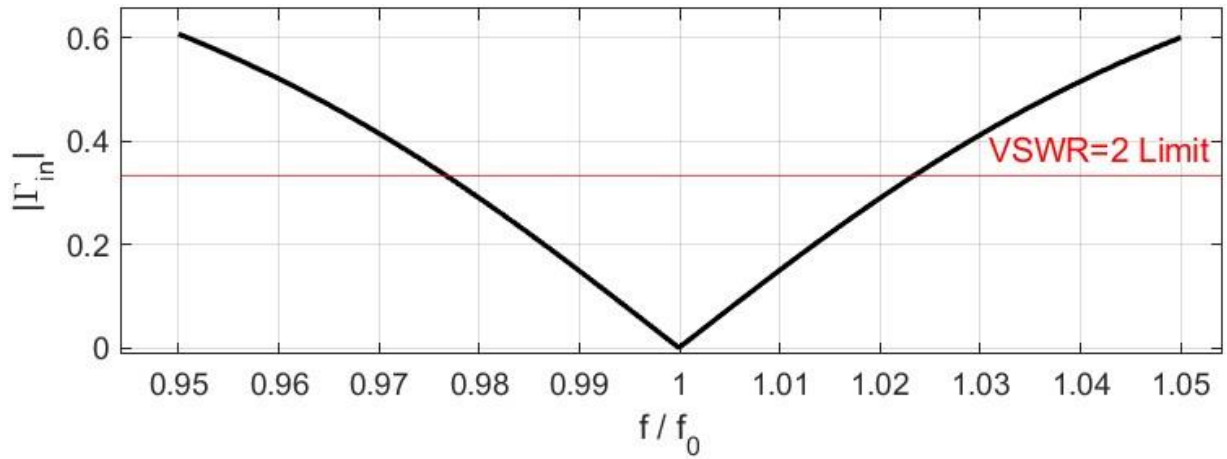


Figure 3: Reflection coefficient magnitude & Operational bandwidth analysis

Comments: The real and imaginary parts of Z_{in} were plotted that showed a perfect match at the centre (design) frequency. The absolute bandwidth was 145MHz, and the respective relative bandwidth was 2.5% of the design frequency for $VSWR \leq 2$.

Part 3:

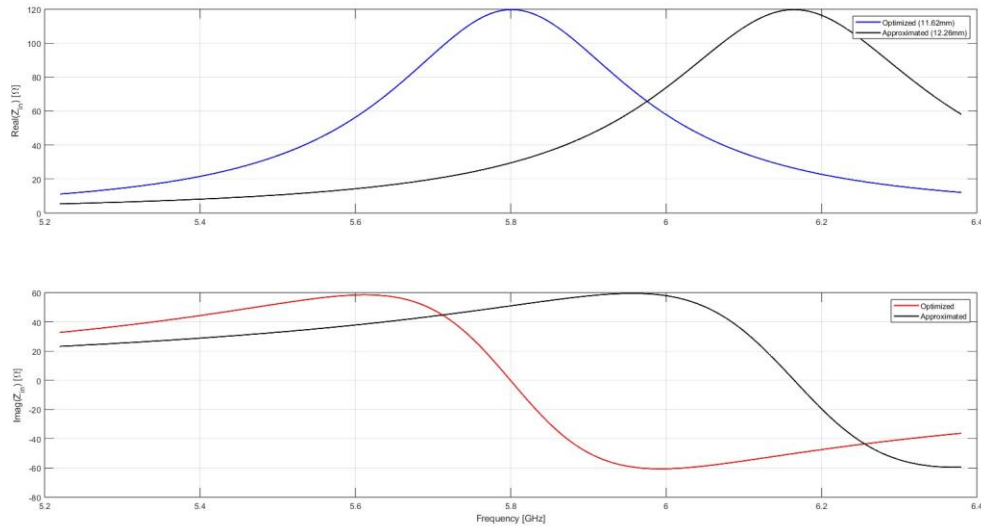


Figure 4: Frequency analysis of approximated vs optimized designs

Comments: The optimized patch peaks at exactly the design frequency while the approximated patch hits the peak at around 6.16GHz. This 360MHz difference could be because the effective permittivity and extra length.

Problem 3

Problem 3.

Using the approximated model to find the position of the probe.

$$\frac{y_{TL}}{\text{Left}} = \frac{1}{\pi} \cos^{-1} \left(\sqrt{\frac{R_{in}}{R_{rad}(0)}} \right)$$

$$\Rightarrow \frac{1}{\pi} \cos^{-1} \left(\sqrt{\frac{50}{119.66}} \right) = 0.276$$

$$y_{TL} = 0.276 \times 1.28 = 0.353 \text{ cm}$$

$$y = y_{TL} - \Delta L \Rightarrow 0.353 - 0.071$$

$$y = 0.282 \text{ cm}$$